



Maharashtra State Board of Technical Education, Mumbai
NEP - 2020 Compliant K (Regular Diploma) / F (Diploma for Working Professionals) / I Scheme Curriculum

K (Regular Diploma) / F (Diploma for Working Professionals) / I Scheme Curriculum Search

Scheme

K (Regular Diploma)

Programme / Course

CO - Diploma In Computer Engineering

Note : Please Click on Course Name / Course Code to View Curriculum

SEMESTER : FIRST																							Scheme Pri					
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme						Credits	Assessment Scheme															
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week	Paper Duration (hrs.)		Theory				Based on LI. & TL				Based on Self Learning		T	M				
						CL	TL	LL					FA-TH	SA-TH	Total	Practical				SLA								
																FA-PR	SA-PR	SLA										
(All Compulsory)																												
1	BASIC MATHEMATICS	BMS	AEC	311302	6	4	2	-	2	8	4	3	30	70	100	40	-	-	-	-	25	10						
2	COMMUNICATION SKILLS (ENGLISH)	ENG	AEC	311303	0	3	-	2	1	6	3	3	30	70	100	40	25	10	-	-	25	10						
3	BASIC SCIENCE	PHYSICS	BSC	DSC	311305	4	2	-	2	1	10	5	1.5	30	70*#	100	40	25	10	25@	10	25	20					
		CHEMISTRY					2	-	2	1								25	10	25@	10	25						
4	FUNDAMENTALS OF ICT	ICT	SEC	311001	0	1	-	2	1	4	2	-	-	-	-	-	25	10	25@	10	25	10						
5	ENGINEERING WORKSHOP PRACTICE (COMPUTER GROUP)	WPC	SEC	311002	0	-	-	4	-	4	2	-	-	-	-	-	50	20	50@	20	-	-						
6	YOGA AND MEDITATION	YAM	VEC	311003	1	-	-	1	1	2	1	-	-	-	-	-	25	10	-	-	25	10						
7	ENGINEERING GRAPHICS (Electronics, Computer and allied branches)	EGP	DSC	311008	2	2	-	4	-	6	3	-	-	-	-	-	50	20	50@	20	-	-						
Total					13	14	2	17	7		20		90	210	300		225		175		150							
Abbreviations : CL- Classroom Learning, TL- Tutorial Learning, LL- Laboratory Learning, FA- Formative Assessment, SA- Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @S Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC), Discipline Specific Elective (DSE), Value Education Course (VEC), Intern./Apprenti./Project/Community (INP), Ability/Enhancement Course (AEC), Skill Enhancement Course (SEC), Generic/Elective (GE) ♦ For the course Basic Science - candidate will have to appear for pre-examination of both physics & chemistry. If absent in any one section (physics / chemistry) student will be declared as absent & fail for the course and marks will not be processed or carried forward.																												

Note : Please Click on Course Name / Course Code to View Curriculum

SEMESTER : SECOND																			Scheme Print								
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme															Total Marks
						Actual Contact Hrs./Week				Self Learning (Activity/ Assignment /Micro Project)		Notional Learning Hrs /Week	Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning						
						CL	TL	LL	FA-TH					SA-TH	Total	Practical				SLA							
																FA-PR	SA-PR	SLA	SLA								
(All Compulsory)																											
1	APPLIED MATHEMATICS	AMS	AEC	312301	2	3	1	-	-	4	2	3	30	70	100	40	-	-	-	-	-	-	100				
2	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING	BEE	AEC	312302	0	4	-	4	2	10	5	1.5	30	70*#	100	40	50	20	50@	20	50	20	250				
3	PROGRAMMING IN C	PIC	AEC	312303	0	4	1	4	1	10	5	3	30	70	100	40	50	20	50#	20	25	10	225				
4	LINUX BASICS	BLP	DSC	312001	0	2	-	2	-	4	2	-	-	-	-	25	10	25@	10	-	-	50					
5	PROFESSIONAL COMMUNICATION	PCO	SEC	312002	0	-	-	2	-	2	1	-	-	-	-	25	10	25@	10	-	-	50					
6	SOCIAL AND LIFE SKILLS	SFS	VEC	312003		-	-	-	2	2	1	-	-	-	-	-	-	-	-	50	20	50					
7	WEB PAGE DESIGNING	WPD	SEC	312004		2	-	4	2	8	4	-	-	-	-	50	20	50@	20	25	10	125					
Total					2	15	2	16	7		20		90	210	300		200	200	150			850					
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @S Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project/Community (INP) , Ability Enhancement Course (AEC) , Skill Enhancement Course (SEC) , Generic Elective (GE)																											

Note : Please Click on Course Name / Course Code to View Curriculum

FIRST YEAR EXIT SCHEME																							Scheme Print			
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme														Total Marks
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning						
						CL	TL	LL					Practical			SLA										
													FA-TH	SA-TH	Total	FA-PR		SA-PR		SLA						
																Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
(All Compulsory)																										
1	WEB PAGE DESIGNING	WPD	SEC	312004		2	-	4	2	8	4	-	-	-	-	50	20	50@	20	25	10	125				
2	EXIT- INTERNSHIP01	EI1	INP	311011		-	-	-	-	0	4	-	-	-	-	25	10	25@	10	-	-	50				
Total						0	2	0	4	2	8	8		0	0	0		75		75		25	175			
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA-Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project/Community (INP) , Ability Enhancement Course (AEC) , Skill Enhancement Course (SEC) , Generic Elective (GE) Student who is exiting after first year will be awarded "Certificate of Vocation" provided following conditions is fulfilled : - Student must have passed all the courses of 1 st & 2 nd semester & completes the exit internship of 4 weeks.																										

Note : Please Click on Course Name / Course Code to View Curriculum

Scheme Print

SEMESTER : THIRD

Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme														Total Marks
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory				Based on LL & TL				Based on Self Learning					
						CL	TL	LL					FA-TH	SA-TH	Total	FA-PR	SA-PR	SLA								
																				Practical						
																				Max	Max	Max	Min	Max	Min	
(All Compulsory)																										
1	DATA STRUCTURE USING C	DSU	DSC	313301	0	3	1	4	-	8	4	3	30	70	100	40	50	20	25#	10	-	-	175			
2	DATABASE MANAGEMENT SYSTEM	DMS	DSC	313302	0	3	1	4	2	10	5	3	30	70	100	40	50	20	25#	10	25	10	200			
3	DIGITAL TECHNIQUES	DTE	DSC	313303	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175			
4	OBJECT ORIENTED PROGRAMMING USING C++	OOP	SEC	313304	0	3	2	4	1	10	5	3	30	70	100	40	50	20	25@	10	25	10	200			
5	COMPUTER GRAPHICS	CGR	DSC	313001	0	1	-	2	1	4	2	-	-	-	-	-	25	10	-	-	25	10	50			
6	ESSENCE OF INDIAN CONSTITUTION	EIC	VEC	313002	0	1			1	2	1	-	-	-	-	-	-	-	-	-	50	20	50			
Total					0	14	4	16	6		20		120	280	400		200		100		150		850			
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA-Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @S Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project/Community (INP) , Ability Enhancement Course (AEC) , Skill Enhancement Course (SEC) , Generic Elective (GE)																										

Note : Please Click on Course Name / Course Code to View Curriculum

SEMESTER : FOURTH																			Scheme Print						
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme													
						Actual Contact Hrs./Week		Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week	Paper Duration (hrs.)		Theory				Based on LL & TL				Based on Self Learning		Total Marks			
						CL	TL					FA-TH	SA-TH	Total	Practical		SLA								
															FA-PR	SA-PR	Max	Min	Max	Min					
(All Compulsory)																									
1	ENVIRONMENTAL EDUCATION AND SUSTAINABILITY	EES	VEC	314301	2	3	-	-	1	4	2	1.5	30	70*#	100	40	-	-	-	-	25	10	125		
2	JAVA PROGRAMMING	JPR	AEC	314317	0	4	-	4	2	10	5	3	30	70	100	40	25	10	50#	20	25	10	200		
3	DATA COMMUNICATION AND COMPUTER NETWORK	DCN	DSC	314318	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175		
4	MICROPROCESSOR PROGRAMMING	MIC	DSC	314321	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25@	10	25	10	175		
5	PYTHON PROGRAMMING	PWP	AEC	314004	0	2	-	4	-	6	3	-	-	-	-	-	50	20	50#	20	-	-	100		
6	UI/UX DESIGN	UID	SEC	314005	0	1	-	4	1	6	3	-	-	-	-	-	25	10	25@	10	25	10	75		
Total					2	16	0	18	6		20		120	280	400		150		175		125		850		
Abbreviations : CL- Classroom Learning, TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA-Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination, @S Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self Learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project/Community (INP) , Ability Enhancement Course (AEC) , Skill Enhancement Course (SEC) , Generic Elective (GE)																									

Note : Please Click on Course Name / Course Code to View Curriculum

Scheme Print

SECOND YEAR EXIT SCHEME																									
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme				Credits	Assessment Scheme														Total Marks
						Actual Contact Hrs./Week		Self Learning (Activity/Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory				Based on L.L. & TL				Based on Self Learning					
						CL	TL					FA-TH	SA-TH	Total	Practical		SLA								
															FA+PR	SA+PR	Max	Min	Max	Min					
General (Any - One)																									
1	EXIT INDUSTRIAL TRAINING (Full Time)	EXIT	INP	312021		-	-	-	-	0	4	-	-	-	-	-	-	50@	20	-	-	50			
	SECOND YEAR EXIT COURSE (Online)	SYEC	SEC	312022		-	-	-	-	0	4	-	-	-	-	-	-	50@	20	-	-	50			
Total						0	0	0	0	0	4		0	0	0		0	50		0		50			
Abbreviations : CL- Classroom Learning, TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination, @S Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project/Community (INP) , Ability Enhancement Course (AEC) , Skill Enhancement Course (SEC) , Generic/Elective (GE) Student who is exiting after first year will be awarded "Certificate of Vocation" provided following conditions is fulfilled : - Student must have passed all the courses of 1 st & 2 nd semester & completes the exit internship of 4 weeks.																									

Note : Please Click on Course Name / Course Code to View Curriculum

SEMESTER : FIFTH																							Scheme Print			
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme						Credits	Assessment Scheme													
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	National Learning Hrs /Week	Paper Duration (hrs.)		Theory				Based on LL & TL				Based on Self Learning		Total Mar			
						CL	TL	LL					FA-TH	SA-TH	Total	Practical				SLA						
																FA-PR	SA-PR	SLA								
(All Compulsory)																										
1	OPERATING SYSTEM	OSY	DSC	315319	0	5	-	2	2	9	3	3	30	70	100	40	25	10	25@	10	25	10	175			
2	SOFTWARE ENGINEERING	STE	DSC	315323	0	4	-	4	1	9	3	3	30	70	100	40	25	10	25@	10	25	10	175			
3	ENTREPRENEURSHIP DEVELOPMENT AND STARTUPS	ENDS	AEC	315002		1	-	2	-	3	1	-	-	-	-	50	20	25@	10	-	-	75				
4	SEMINAR AND PROJECT INITIATION COURSE	SPI	AEC	315003		-	-	1	2	3	1	-	-	-	-	25	10	25@	10	25	10	75				
5	INTERNSHIP(12 WEEKS)	ITR	INP	315004		-	-	-	-	36 - 40	10	-	-	-	-	100	40	100#	40	-	-	200				
Elective I (Any -One)																										
6	ADVANCE COMPUTER NETWORK	ACN	DSE	315321	0	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150			
	CLOUD COMPUTING	CLC	DSE	315325	0	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150			
	DATA ANALYTICS	DAN	DSE	315326	0	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150			
Total						0	14	0	11	5		20		90	210	300		250	225		75		850			
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Note : Please Click on Course Name / Course Code to View Curriculum

SEMESTER : SIXTH																				Scheme Print						
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme						Credits	Assessment Scheme													
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week	Paper Duration (hrs.)					Based on LL & TL				Based on Self Learning		Total Marks				
						CL	TL	LL					FA-TH	SA-TH	Total	Practical				SLA						
																Max	Min	Max	Min		Max		Min			
(All Compulsory)																										
1	MANAGEMENT	MAN	AEC	315301	1	3	-	-	1	4	2	1.5	30	70*#	100	40	-	-	-	-	25	10	125			
2	EMERGING TRENDS IN COMPUTER ENGINEERING AND INFORMATION TECHNOLOGY	ETI	DSC	316313	0	3	-	-	1	4	2	1.5	30	70*#	100	40	-	-	-	-	25	10	125			
3	SOFTWARE TESTING	SFT	DSC	316314		3	-	4	1	8	4	3	30	70	100	40	25	10	-	-	25	10	150			
4	CLIENT SIDE SCRIPTING	CSS	AEC	316005	0	2	-	4	-	6	3	-	-	-	-	-	25	10	25@	10	-	-	50			
5	MOBILE APPLICATION DEVELOPMENT	MAD	DSC	316006	0	2	-	4	2	8	4	-	-	-	-	-	25	10	25#	10	25	10	75			
6	CAPSTONE PROJECT	CPE	INP	316004		-	-	2	2	4	2	-	-	-	-	-	50	20	50#	20	50	20	150			
Elective 2 (Any - One)																										
7	DIGITAL FORENSIC AND HACKING TECHNIQUES	DFH	DSE	316315	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175			
	MACHINE LEARNING	MAL	DSE	316316	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175			
	NETWORK AND INFORMATION SECURITY	NIS	DSE	316317	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175			
Total					1	16	0	16	8		20		120	280	400		150		125		175		850			
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